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Engineer Module Documentation

Dependencies

```
using Application.Helpers;
using Application.Interfaces.Admin;
using Application.Services.Admin;
using AutoMapper;
using Domain.DTOS;
using Domain.Entities;
using Domain.Resources;
using FougeraClub.Areas.Admin.ViewModels.Engineer;
using FougeraClub.Attributes;
using FougeraClub.Helpers;
using Infrastructure.Repositories.InterfacesDB;
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Mvc.Rendering;
```

Helper Classes Used

Helper	Purpose
SessionHelper	Retrieves current language (used for multilingual UI)
SelectListHelper	Generates dropdown lists for nationalities
FileHelper	Validates uploaded images (type & size)
PaginatedList<T>	Handles pagination logic for large datasets
ExcelStaticReport	Generates Excel reports for export

Attributes

Attribute	Description
[AdminAuthorize]	Restricts access to Admin users only
[Area("Admin")]	Declares this controller belongs to the Admin area
[IgnoreAction]	Marks actions that should be excluded from permission scanning

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Views Used

Action	View Name	Partial View
Index	Index.cshtml	_ListPartial.cshtml
AddEdit	AddEdit.cshtml	—
Details	Details.cshtml	—
Print	Print.cshtml	—
PrintDetails	PrintDetails.cshtml	—

Summary of Responsibilities

Feature	Description
Listing & Filtering	Displays paginated engineers with filtering options
CRUD Operations	Add, Edit, Delete engineers
File Upload	Validates and uploads profile image
Localization	Adjusts displayed data according to current language
Reports	Generates printable and Excel reports
Dropdown Management	Dynamically populates Nationalities and Graduation Years

Controller Summary

Method	Type	Description
Index	GET	List engineers with filtering & pagination
AddEdit	GET	Display form for Add/Edit
AddEdit	POST	Save new or updated engineer
Delete	POST	Delete engineer
Print	GET	Generate printable list
createExcelReport_Download	GET	Generate Excel file
Details	GET	View engineer details
PrintDetails	GET	Printable detail page

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Engineer Model Properties

Property	Type	Description	Constraints / Notes
Id	int	Unique identifier for the engineer	Primary key 
FullName	string?	Engineer's full name	MaxLength(100)
Code	int?	Engineer-specific internal code	Optional
Specialization	string?	Engineering field (e.g., Civil, Electrical)	MaxLength(100)
GraduationYear	int?	Year of graduation	Range(1900, 2100)
DateOfBirth	DateTime?	Date of birth	Stored as date in DB 
Position	string?	Job position or title	MaxLength(100)
NationalityId	int?	Foreign key referencing Nationality	Optional
Nationality	Nationality?	Navigation to nationality entity	Virtual navigation property
WorkAddress	string?	Address of current workplace	MaxLength(200)
Address	string?	Home or personal address	MaxLength(200)
PhoneNumber	string?	Contact phone number	Phone validation + MaxLength(20)
Email	string?	Email address	EmailAddress + MaxLength(100)
Notes	string?	Optional notes or comments	No constraint
ProfileImagePath	string?	Path to profile image	MaxLength(255)
PassportNumber	string?	Passport number	MaxLength(50)
PassportExpiryDate	DateOnly?	Passport expiration date	Stored as date 
IdNumber	string?	Government-issued ID number	MaxLength(50)
IdExpiryDate	DateOnly?	National ID expiration date	Stored as date 
IdReleaseLocation	string?	Place where the ID was issued	MaxLength(50)
Salary	decimal?	Monthly or yearly salary	Range(0, 1,000,000) 
BankAccountNo	string?	Bank account number	MaxLength(50)
BankName	string?	Name of the bank	MaxLength(50)

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Engineer Relationships

Property	Type	Description
Nationality	Nationality?	Linked nationality entity (optional)

Service Methods

Method	Description
GetAllAsync()	Gets all engineers with nationalities
GetByIdAsync(id)	Fetches specific engineer + nationality
AddAsync(entity, file)	Saves image (if provided), stores engineer
UpdateAsync(entity, file)	Updates details, replaces image
DeleteAsync(id)	Deletes record and related image
SaveImageAsync(file)	Saves image to /uploads/engineers
DeleteImageFile(path)	Deletes image from disk
GenerateNewCode()	Generates sequential Code